

PROJECT BRIEFING

TalentFinder

Automated candidate sourcing, scoring and ranking

A recruiter types the requirements for a role. The system finds candidate profiles, extracts each into structured data, scores every candidate on four weighted dimensions, discards the ones failing a hard requirement, and returns a ranked shortlist with a written justification for each person.

This document is for understanding the whole system well enough to explain and defend it: what it does, how it is built, what it costs, where it is fragile, and which decisions are still open.

STATUS

Working · proven on a live search

QUALITY

419 tests passing

CODEBASE

Branch `main` , pushed

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HOW TO USE THIS DOCUMENT

Sections 1–2 are the business story: read these if you have five minutes. Sections 3–7 are how it is built, in the order you would explain it at a whiteboard. Sections 8–14 are what an owner cares about — quality, cost, risk, and what happens next.

1. Executive summary

THE PITCH

Recruiters spend most of their sourcing time reading profiles that were never going to fit. This automates that first pass and returns a ranked shortlist where every candidate carries a match percentage, a breakdown of why, and the specific things they have and lack.

4

SCORING DIMENSIONS

6 / 6

LIVE SEARCH, CORRECT

33 / 33

EMPLOYER FILTER
CORRECT

419

AUTOMATED TESTS

It has been proven on a real search

A recruiter asked for external audit managers at the Big Four with a professional qualification and five years' experience, and received six:

100	Hisham Sabry, ACCA, ESAA	Audit Manager	EY
100	Marwan Mohamed, CPA	Audit Manager	KPMG
95	Mohamed Fathy Bakr, ACCA, FESAA	Audit Manager	EY
92	Mohamed Elsherif, ACCA, CA, DipIFR	Audit Manager	EY
92	Belal Mansour, ESAA, DipIFR	Assistant Manager	Deloitte
85	Nancy Hamdy, ACCA	Audit Manager	EY

The three things that matter most

- 1. The valuable part is the scoring engine, not the scraper.** The data source is a swappable component. Matching, ranking and explanation work identically whether profiles come from LinkedIn, a paid vendor, or your own applicant database. Adding a source is one class with two methods. The asset survives any decision about where data comes from.
- 2. Every score is explainable and reproducible.** No black box. A candidate scoring 92 has a stored breakdown showing exactly which component contributed what, and the rules are versioned, so a score from six months ago still means something. This matters commercially and legally — automated decisions about people need to be defensible.
- 3. The LinkedIn path is the riskiest component.** It breaches LinkedIn's terms and gets accounts restricted — twice so far. The code takes that seriously: a hard hourly cap, manual sign-in, no stored passwords, and an account-rotation path. But no amount of engineering makes it safe, and section 14 is the honest recommendation.

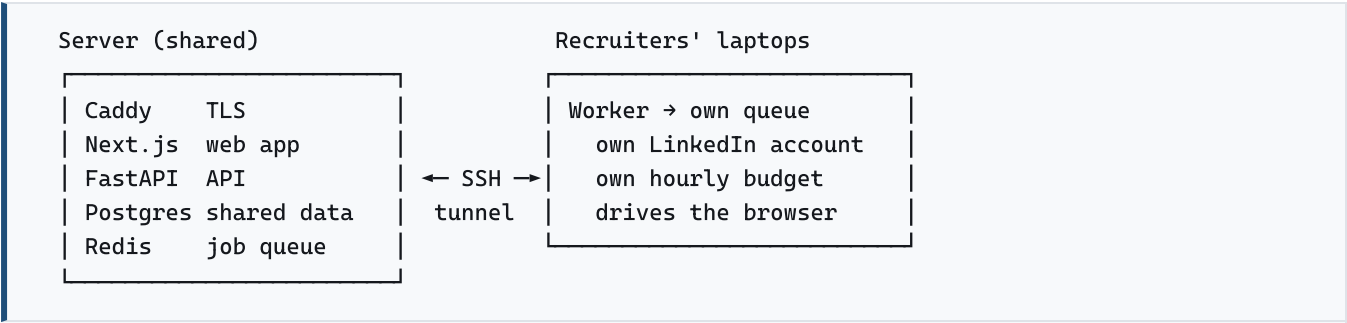
2. What a recruiter actually does

1. **Describe the role.** Job title, optional keywords, qualifications, minimum experience, location, employers, graduation-year window.
2. **The search runs in the background.** A browser opens on their own machine; the first run of the day asks them to sign in to LinkedIn by hand. Progress streams live.
3. **Read the ranked results.** Each candidate shows a match score, the four-part breakdown, matched and missing keywords, and plain-English reasons.
4. **Save and export.** Shortlist to a saved list, or straight to CSV.

THE ONE FIELD PEOPLE GET WRONG

Critical skills are pass/fail, and separate entries mean *all* of them. Three entries — CPA, ACCA, ESAA — requires a candidate to hold all three, which is nobody. To accept any of several qualifications, put them in one entry: CPA / ACCA / ESAA . The form now shows the logic as it is typed and warns when an unintended "and" has been built.

3. Architecture



Why the split. Sign-in and CAPTCHAs are cleared by a person in a visible browser window. A server has no screen, so the scraper cannot live there. Everything shared — the database, the web app, the queue — does.

Why per-recruiter queues. A search runs only on its owner's laptop. Celery's default hands any job to any free worker, which here would load one recruiter's LinkedIn session into a colleague's browser on a different IP — unusable, and close to the textbook signal of a stolen account.

PIECE	WHY
FastAPI	Typed, async, automatic API documentation.
Celery + Redis	Searches take minutes; they cannot run inside a web request.
PostgreSQL	Shared candidate pool, so two recruiters never pay twice to collect the same person.
Next.js	Server rendering plus a live progress stream.
Playwright	Drives a real browser — the only way to read a site built for humans.
Supabase	Authentication only. No candidate data goes there.

4. The search pipeline

Six stages. If you can narrate these, you can answer almost any question about behaviour.

STAGE	WHAT HAPPENS
1. Search	Only the job title goes in the search box. Employer and location are applied through LinkedIn's own filters, which restrict server-side so unqualified people are never opened.
2. Pre-filter	Deliberately timid — discards a card only when it clearly contradicts a stated requirement. A false negative here is invisible and unrecoverable; a false positive costs one page load.
3. Fetch	Checks the cache first. A profile collected in the last week is reused and costs nothing — no page load, no budget, no exposure.
4. Score	Four weighted components. See section 5.
5. Filter	Hard requirements: qualifications, employer, experience, career stage, score threshold.
6. Store	Ranked and written with the score breakdown, the reasons, and the version of the rules used.

TWO BEHAVIOURS WORTH KNOWING BY HEART

Partial results beat failed searches. If the hourly budget runs out mid-search, or the account is restricted, or the machine dies — everything collected so far is kept and stored. Because profiles are cached, re-running picks up where it left off rather than paying again.

One bad profile does not kill a search, but several in a row stop it. A single unreadable profile is skipped; three consecutive failures mean something systemic — an expired session, a layout change — and continuing would spend the rest of an hourly budget repeating one failure.

5. The matching engine

Everything else is plumbing. This is the part that produces the number a recruiter acts on.

COMPONENT	WEIGHT	HOW IT IS COMPUTED
Title	30%	Split into <i>what the job is</i> and <i>how senior it is</i> , scored separately. The domain is a gate: no evidence of the work scores zero, however senior. Judged across the whole career.
Keywords	30%	Fraction found in the candidate's name, headline, job titles, skills and certifications. Tolerates word forms — "vendor management" matches <i>Vendor Manager</i> .
Experience	30%	Total years against the minimum, capped. Overlapping roles merged and career gaps excluded, so nobody is credited twice for two jobs at once.
Location	10%	A preference, not a filter, unless the strict switch is on.

Any component left blank is dropped and its weight redistributed. Nobody is marked down for failing a requirement that was never set — which is what makes the optional fields genuinely optional.

What v4 fixed, and why it took real data to find

THREE DEFECTS, ALL FOUND BY MEASURING RATHER THAN READING THE CODE

- 1. "Manager" is not a job.** A search for *external audit manager* returned Finance Managers, because both share a word that describes a *level* and appears in every hierarchy. Domain and seniority are now separate, and the domain is a gate.
- 2. A self-declared skill tag is not a career.** Six candidates cleared the gate on a LinkedIn tag reading "External Audit" with no supporting job history. Skills now count for keyword matching, not for the profession gate.
- 3. At an audit firm, external audit is called "Audit".** The most serious. Demanding the literal word discarded **28 audit managers at the exact four firms the recruiter asked for**. An unqualified audit role now reads as external audit unless the career says "internal" — which internal auditors always do, because theirs is the variant needing a label.

Measured on 237 real profiles

SEARCH	BEFORE	NOW	REJECTED
external audit manager	237	72	70%
financial reporting supervisor	237	49	79%
finance manager	237	91	62%

The most specific request cuts hardest, the broadest cuts least — the pattern that shows the gate discriminates rather than simply rejecting. Full detail in the companion document, *Matching Accuracy, Measured*.

KNOW THIS FOR THE ROOM

CPA is the American qualification. Across the 237 profiles: CPA 42, ACCA 21, DipIFR 16, CIA 8, ESAA 5. For Egypt and Saudi Arabia, CPA / ACCA / ESAA is the honest default and CPA alone is a US filter.

6. Telling the truth about failure

This is the part that separates a tool recruiters keep using from one they abandon, and it was the largest single piece of recent work.

A search used to have two outcomes: it finished, or it crashed. Anything in between — a filter that never reached LinkedIn, a session that expired half-way, a results page that did not render — *finished*, and produced the same message:

"No candidates matched. Try lowering the minimum score or experience."

That sentence blames the recruiter's criteria for a technical failure. A recruiter told it twice concludes the market is empty, or the tool does not work, and they are right to.

WHAT HAPPENED	WHAT THE RECRUITER IS TOLD NOW
Nobody matched	Which filters rejected people and how many — <i>"8x works at a different employer, 2x missing a required qualification"</i>
A filter could not be applied	<i>"These results are not what you asked for"</i> , above the results, plus how to work around it
Hourly limit reached	That it was cut short, that the criteria are not the problem, and when it resumes
Account restricted	That it stopped and that collected work was kept
Their machine died	That it stopped unexpectedly and re-running will not repeat the cost

THE PRINCIPLE

"No mistakes" is not achievable — the data source is someone else's website and it will change without notice. What is achievable, and what is now built: *the app never presents an unreliable answer as a reliable one.*

And when the machine simply vanishes

A worker that dies mid-search used to leave the search saying "running" for ever. A sweep now marks abandoned runs failed, measured from the last progress update, so a slow search is never reaped out from under itself. Job redelivery is disabled deliberately: re-running a search re-opens profiles, re-spending budget that takes an hour to refill, and would collide with the results the first attempt already stored.

7. Where the data comes from

SOURCE	STATUS	NOTES
Demo data	WORKING	Fixtures. No account, no network, no cost. What you demonstrate with.
LinkedIn	WORKING, RISKY	Collected by us with a recruiter's own account. Breaches the terms. Two accounts restricted so far.
Indeed	BUILT, UNPROVEN	Wired and tested, but its extraction cannot be verified without a paid employer account.
Licensed vendor	NOT CHOSEN	The decision in section 14. Removes every risk above at once.

How the scraping is kept survivable

LAYER	WHAT IT DOES
The volume cap	The control that actually matters. A hard hourly ceiling, counted in a rolling window persisted to disk so it survives restarts. Volume is what abuse detection keys on.
Manual sign-in	The app never types a password. A person signs in and clears the CAPTCHA. There are no stored credentials to leak.
Human behaviour	Curved pointer paths, inertial scrolling, irregular pacing with pauses, honeypot checks before clicking.
Fingerprint consistency	One stable machine identity per account, distinct between accounts — so a restriction on one does not link to the next.

THE CEILING, STATED PLAINLY

Throughput scales with recruiters, not with hardware: each account has its own hourly allowance. Five recruiters is five times the rate — and five personal accounts at risk. That is the trade this design makes, and it is the reason section 14 exists.

8. Quality: what 419 tests cover

AREA	WHAT IS PINNED
Scoring and filtering	Every component, the weights, the domain gate, credential alternatives, the employer guarantee.
Experience maths	Overlapping roles, career gaps, a dozen date formats.
The truth-telling	That a degraded run cannot report itself as an ordinary empty one; that repeated failures stop a run.
Failure recovery	Abandoned searches reach a terminal state; a slow one is not reaped.
Data protection	Deletion removes every copy; retention never deletes work in use; one recruiter cannot read another's sourcing.
Readiness	That it fails when a dependency is down — a health check that cannot fail is worse than none.

What is not tested, stated honestly

The scraper's selectors. They depend on LinkedIn's live markup, which no test can pin — a test would only assert that today's HTML looks like today's HTML. This is covered by debug dumps written on failure and by the degraded reporting, not by tests, and it is the part most likely to break.

9. Data protection and retention

The system stores real people's professional histories. Three obligations, all now implemented rather than promised:

OBLIGATION	HOW IT IS MET
Deletion on request	A candidate can be erased: their profile, every search result referencing them, and every recruiter's saved copy. Logged, because the record that would prove the request was honoured is the one being destroyed.
Retention limit	Profiles nothing references are deleted after six months. Previously nothing ever removed a candidate — the cache window only re-scraped them.
Data minimisation	The raw collected payload is discarded once a profile is old enough to be re-fetched rather than reused.
Access control	A recruiter can read candidates from their own searches and saved lists. Profiles are shared internally for efficiency but not readable by id alone.

10. Deployment and operations

The server runs five containers behind TLS: web app, API, Postgres, Redis and a reverse proxy that obtains its own certificate. Database and queue are bound to loopback and reached from recruiters' laptops through an SSH tunnel — nothing but ports 80 and 443 is exposed.

OPERATIONAL CONCERN	HOW IT IS HANDLED
Is it actually working?	A readiness endpoint that checks the database and queue and returns an error when either is down. Liveness is separate and dependency-free, so a blip does not get a healthy process killed.
Setting up a laptop	A preflight script checks Python, the browser, the tunnel, the state directory and the power settings, and prints the fix for each failure.
Diagnosing a bad search	The worker writes a dated log file — the only place a stack trace is captured anywhere in the system.
When something breaks	A runbook starting from the report you will actually get: "my search did nothing", which now has six distinguishable causes.
Backups	A nightly dump with retention, empty-dump detection, and optional off-site copy. It must be installed and rehearsed — an untested backup is an assumption.

11. Cost

ITEM	COST	NOTES
Server	~€5-10/mo	2 vCPU / 4 GB is ample.
Domain	~€10/yr	Or free via a dynamic-DNS subdomain.
Authentication	Free	Free tier covers a company's recruiters comfortably.
LinkedIn data	"Free"	Paid in account risk and unplanned repair work, not money.
Licensed data	To be quoted	The decision in section 14.
AI matching	Optional	Off by default; costs per candidate when on.

Infrastructure is close to free. The real cost is the data, and today it is being paid in a currency that does not appear on an invoice.

12. Legal and compliance

Scraping LinkedIn breaches its User Agreement. Raise this yourself rather than let it be discovered. Three concrete consequences:

1. **Account restriction.** Happened twice. Recoverable only by the account's owner, usually with identity documents.
2. **It becomes the company's breach.** Rolling out company-wide moves this from one person's account to the company, at company direction, using employees' personal accounts.
3. **Personal data obligations.** Collecting professional histories at scale carries them regardless of source. Section 9 covers what is built.

This document is engineering analysis, not legal advice. A lawful basis and jurisdiction-specific obligations should be confirmed with counsel before processing real candidate data at scale.

13. Known limitations

LIMITATION	SEVERITY	DETAIL
Scraping does not scale	HIGH	A fixed hourly ceiling per account, and each account is a person's own. Structural, not a tuning problem.
Extraction will break	HIGH	Depends on LinkedIn's markup. Twice already. Debug dumps make repair quick, but it is unplanned work on someone else's schedule.
The employer filter needs a manual step	MEDIUM	Restricting to named firms works reliably by pasting a LinkedIn URL — ten seconds. Driving the filter panel automatically is unproven, and the app says so when it fails.
No labelled accuracy benchmark	MEDIUM	About an hour of a recruiter's time would produce a defensible single figure. Until then, only the specific measured results can be claimed.
Binary location scoring	MEDIUM	No concept of distance or commuting range — "Berlin" and "Munich" are equally wrong.
Not yet deployed	MEDIUM	Runs on a developer machine. Deployment is documented and needs a server and about an hour.
Accounts burn faster than they rotate	HIGH	Rotation defeats device correlation only. The IP is unchanged, so each replacement is more exposed than the last. A supply problem, not a bug.

14. Where I would take it next

Immediate — days, independent of any strategic decision

- Deploy to a small server and run it for two weeks with one or two recruiters.
- Label 50 candidates to get a defensible accuracy figure.
- Install and *rehearse* the backup restore.

Short term — weeks

- Finish the automatic employer filter, or accept the paste-a-URL step as the supported path.
- Roll out laptop by laptop, each with the preflight check.

Strategic — the decision that unblocks everything

Choose where candidate data comes from

Everything difficult about this system traces to one choice: we collect the data ourselves from a site that does not want us to. That is what caps throughput, what puts recruiters' personal accounts at risk, what makes it a terms breach, and what breaks without warning.

Buying from a licensed provider removes all four at once, and **nothing else about the product changes** — the source was built as a replaceable component from the start, precisely so this decision could be made late and commercially rather than early and technically.

The ask: approval to evaluate providers and a budget range, plus a decision on whether recruiters' personal LinkedIn accounts may be used meanwhile.

Prepared 14 August 2026 against branch `main`. Accuracy figures come from re-running the production code over the 237 candidate profiles in the application database, plus one live LinkedIn search. 419 automated checks passing.